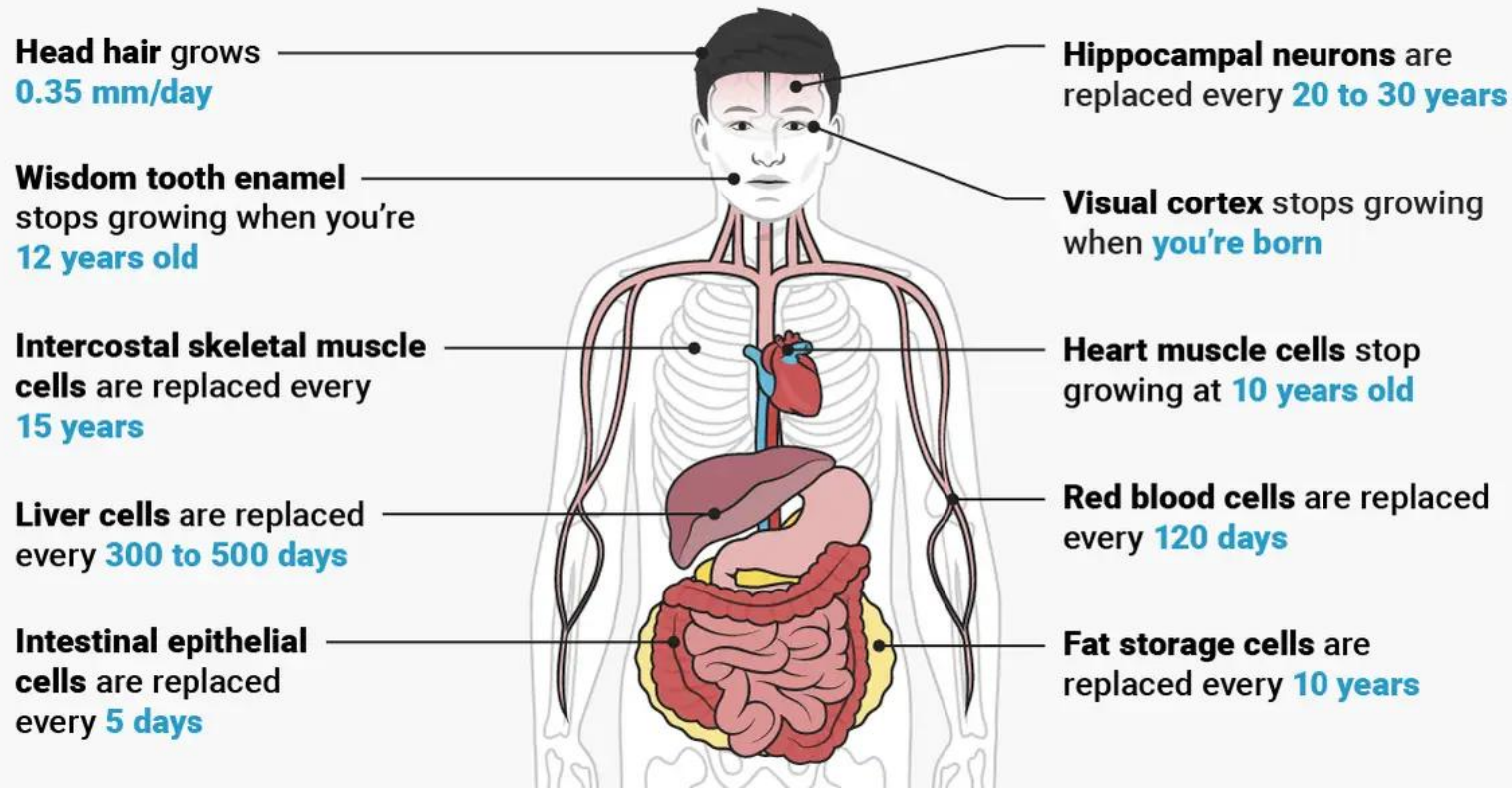


Life Time of Cells in Human Body

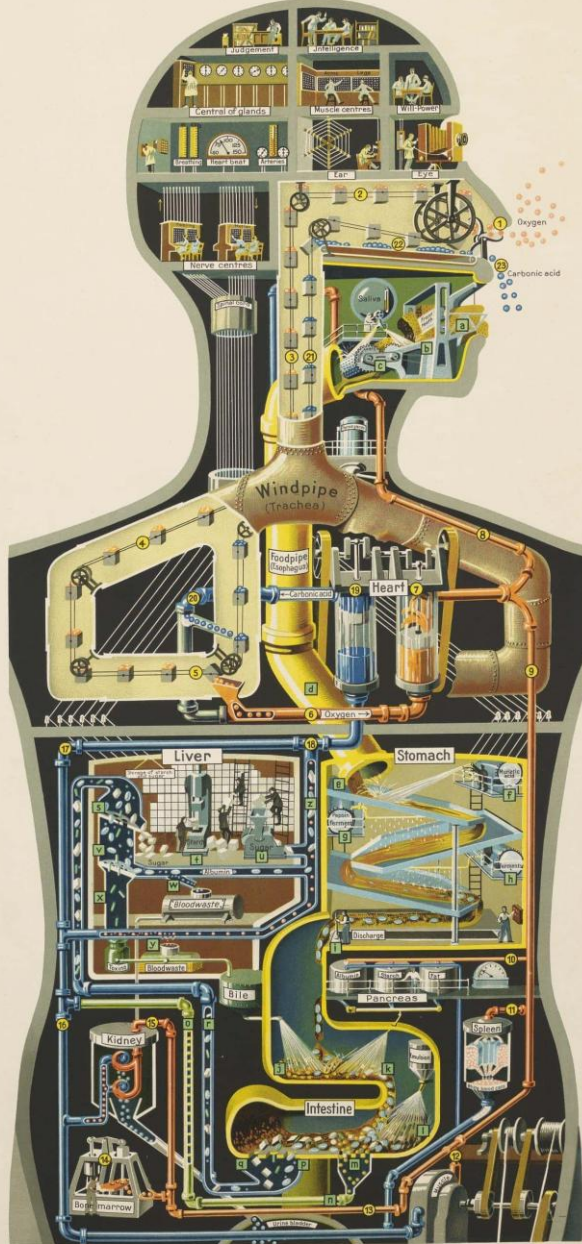
Cell Type	Life Span	Cell Division
Lining of esophagus	2-3 days	Can divide
Lining of small intestine	1-2 days	Can divide
Lining of large intestine	6 days	Can divide
Red blood cells	Less than 120 days	Cannot divide
White blood cells	10 hours to decades	Many do not divide
Smooth muscle (along digestive system)	Long lived	Can divide
Cardiac (heart) muscle	Long lived	Cannot divide
Skeletal muscle	Long lived	Cannot divide
Neuron (nerve cell)	Long lived	Most do not divide

- Highly dynamic, yet efficient
- Endogenous resource utilisation
- Process is the same as that of the nutrients taken from the food



THE HUMAN FACTORY

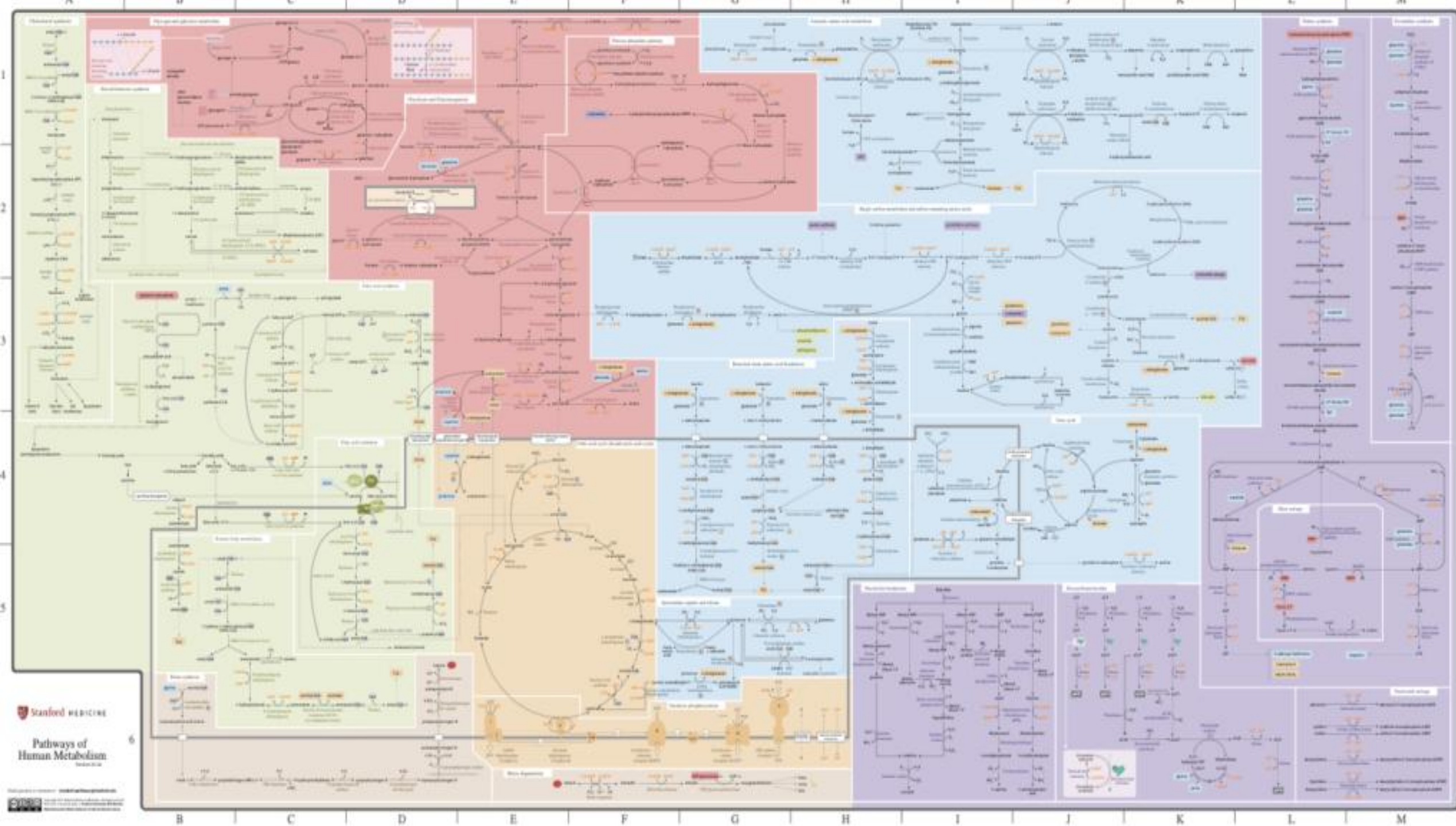
Sketch of the chemical functions of the body



THE HUMAN LIFE

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Printed in Germany

Iconic Fritz Kahn infographic of the Human Factory



Broad Classification of the Utilisation Process

The process by which the absorbed nutrient is utilised by the system (human body) is called **Metabolism**

It is the chemical reactions involved in maintaining the living state of the cells in an organism (sustain life)

Broad classification of metabolic process

Anabolism (synthesis): Uses energy to construct components of the cell – eg. proteins and nucleic acids; building up of tissues and energy stores

Catabolism (degradation): Break down of organic matter; breaking down tissues and energy stores

Both these processes are balancing acts which go on at the same time

Main functions of metabolism

- Convert food into energy for running the cellular processes
- Convert food or fuel to building blocks of the body including proteins, nucleic acids, carbohydrates and lipids
- Elimination of nitrogenous wastes

All these processes are catalysed by enzymes. The process allows the organisms to grow and reproduce, maintain structures and respond to their respective environment

Bird's Eye View of the Utilisation of the Nutrients Derived from Food

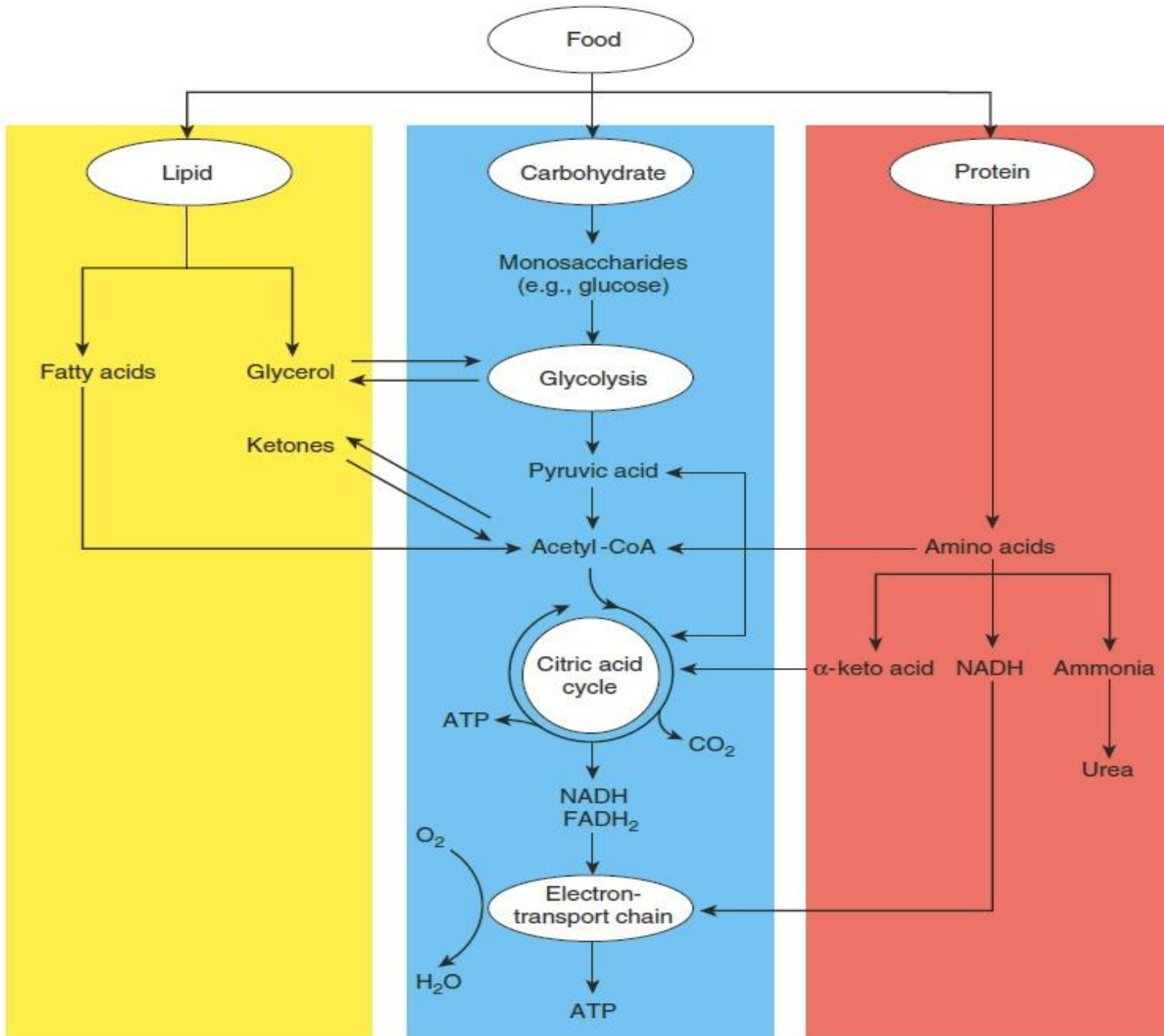
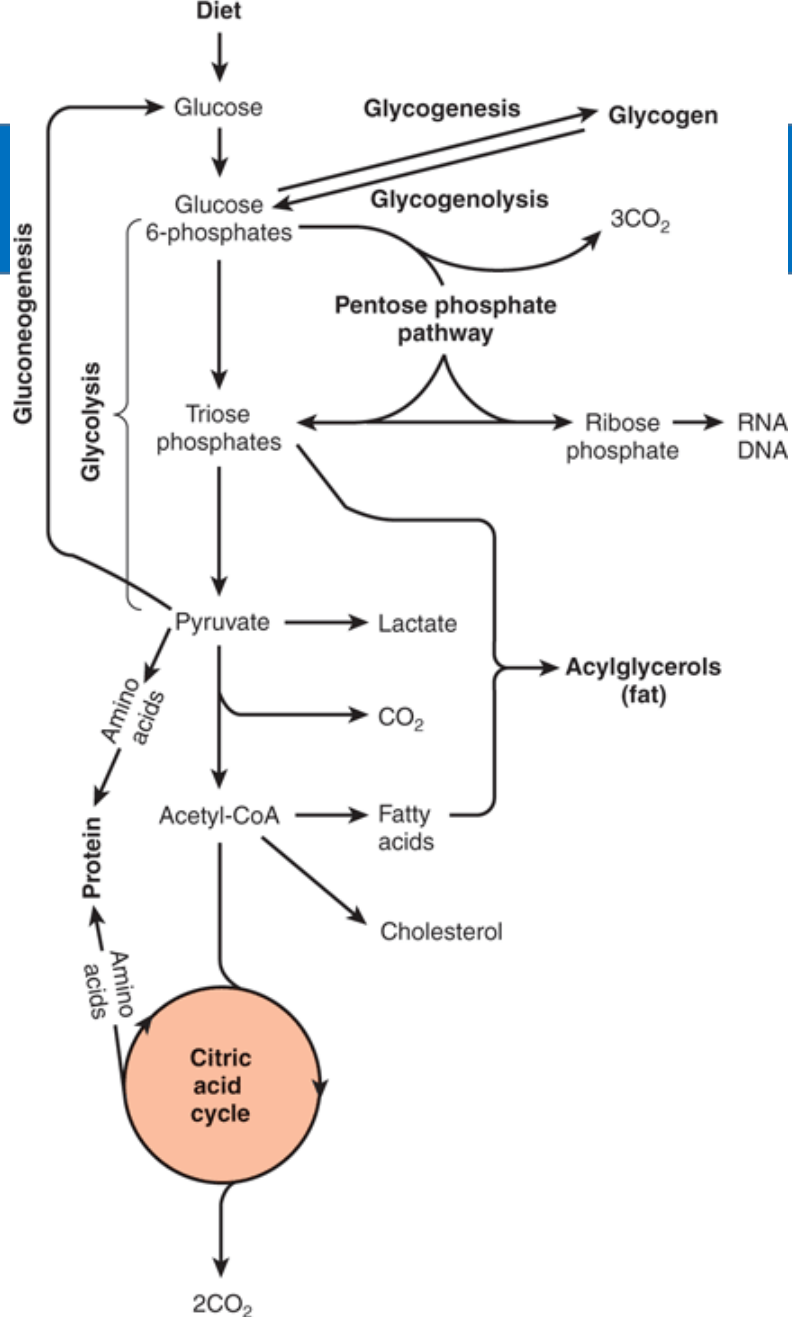
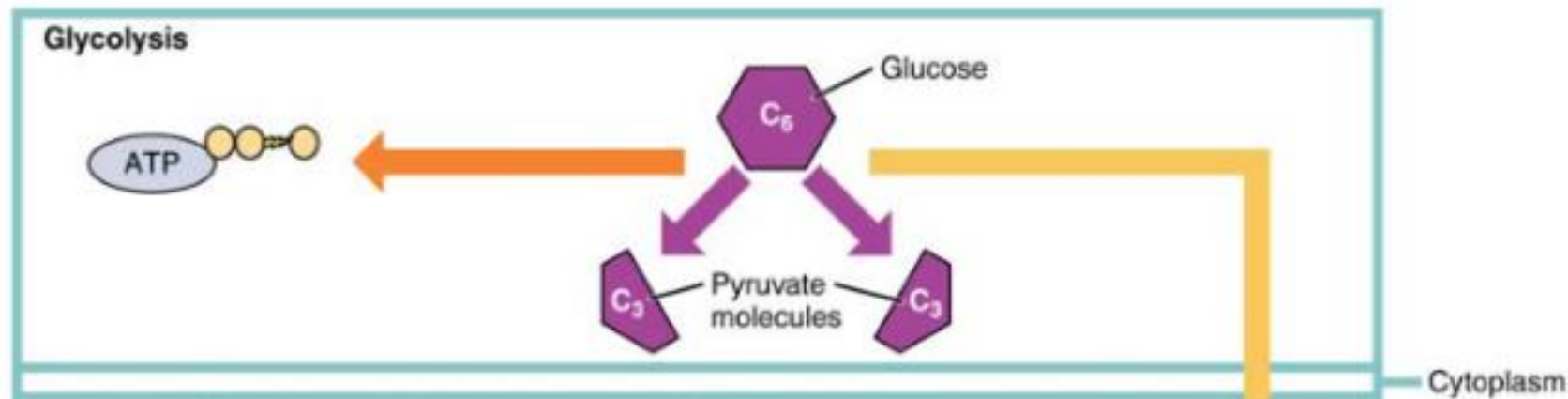


Figure: Overall pathways for metabolism of carbohydrates, proteins and lipids.



Utilisation of Carbohydrates

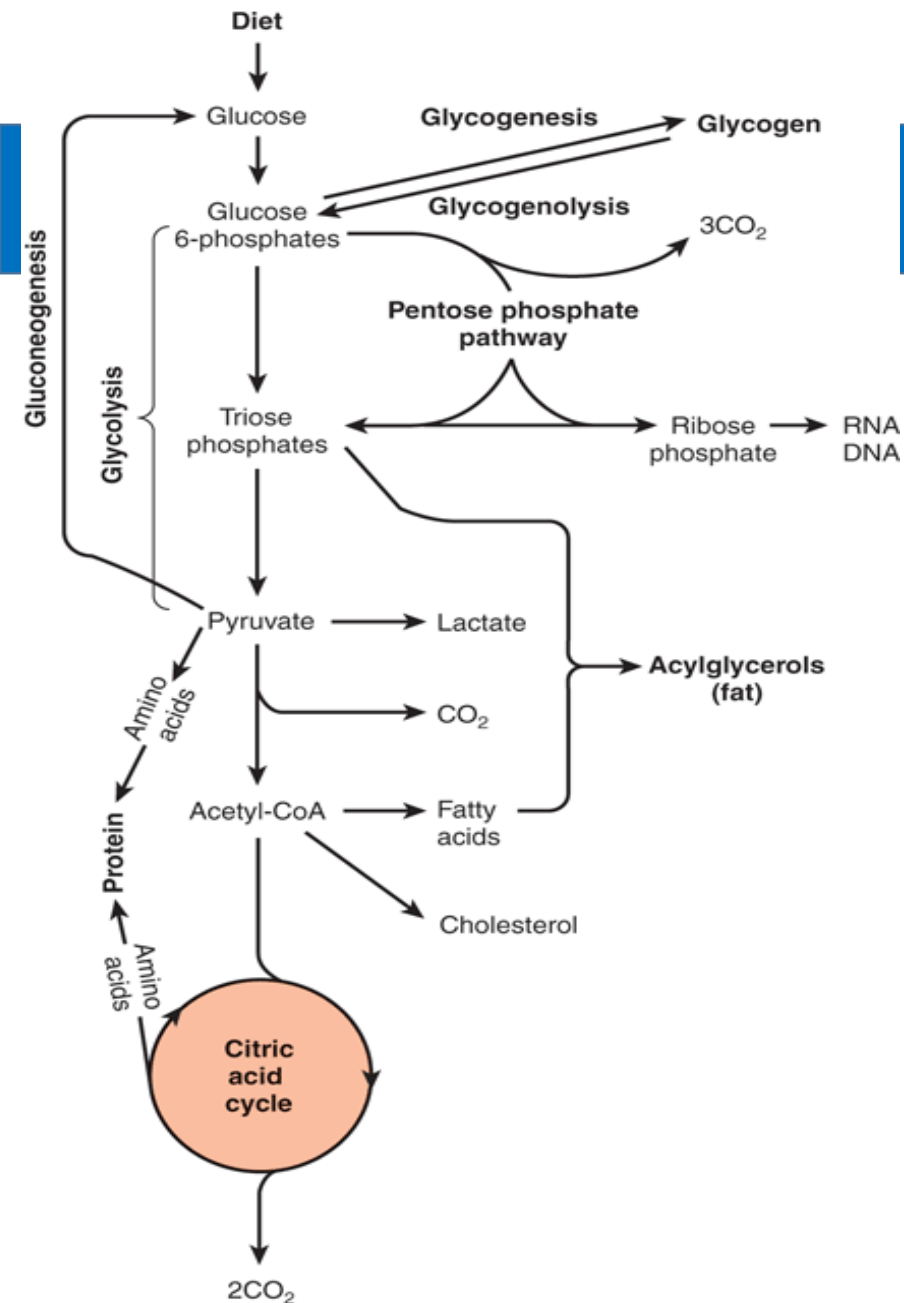
- After absorption, the monosaccharides are transferred to the tissues
- The process of **cellular respiration** begins in the cells, consisting of the following –
- **(1) Glycolysis:** Monosaccharide glucose is oxidised releasing the energy stored in its bonds to produce Adenosine triphosphate (ATP)
- End product of glycolysis is pyruvate ($C_6 \rightarrow 2 \times C_3$) and 2 moles of ATP (from one mole of glucose)
- Intermediates are responsible for the formation of nucleic acids



Utilisation of Carbohydrates

- (2) Anaerobic / aerobic respiration:

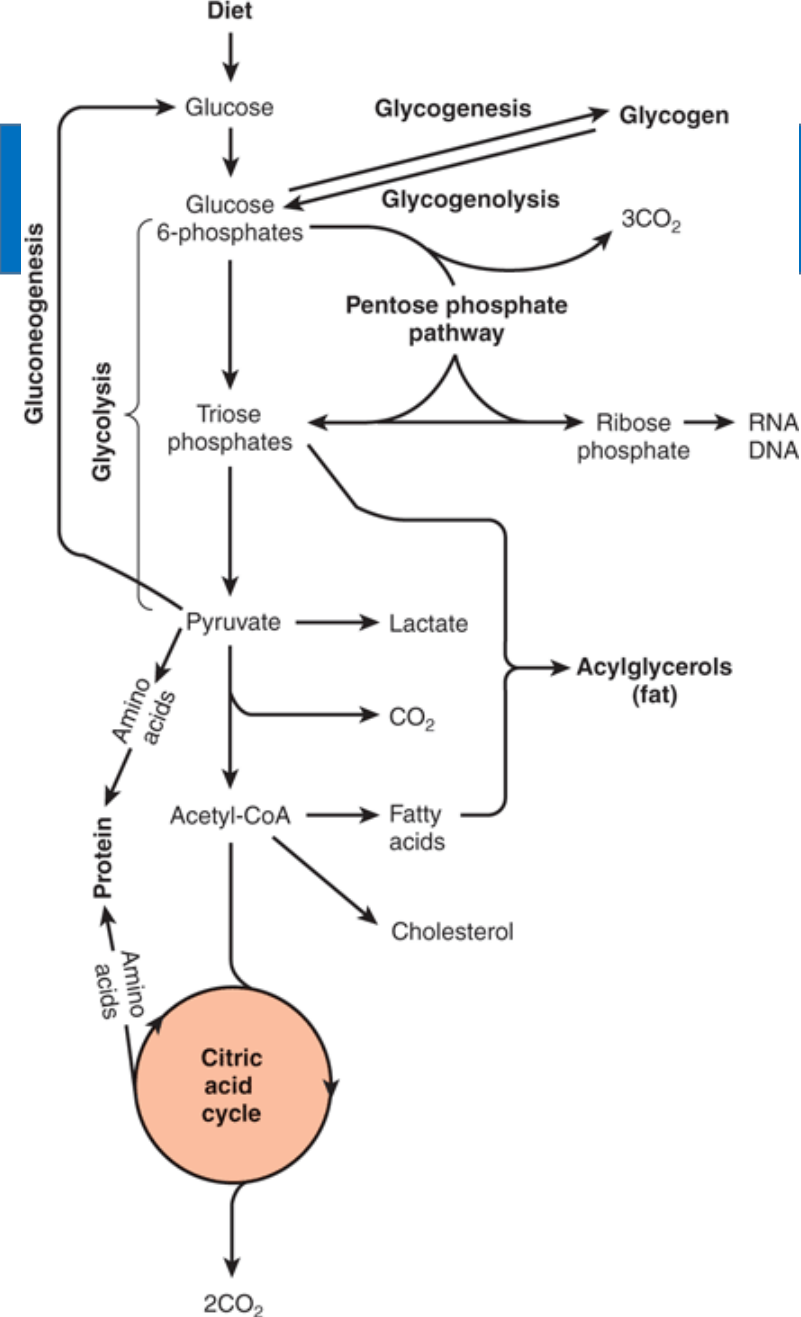
- Anaerobic respiration occurs in the limited presence or absence of oxygen. It occurs when mitochondria is absent or non-functional
- Pyruvate is converted to lactic acid
- Pyruvate concentration need to be maintained lower so that glycolysis continues
- Effective pathway for ATP production in shorter time (eg by erythrocytes – red blood cells) – seconds to minutes
- Lactic acid diffuses to the plasma and then to liver where it is converted back into pyruvate or glucose (Cori cycle)
- Eg. during exercise oxygen supply is limited

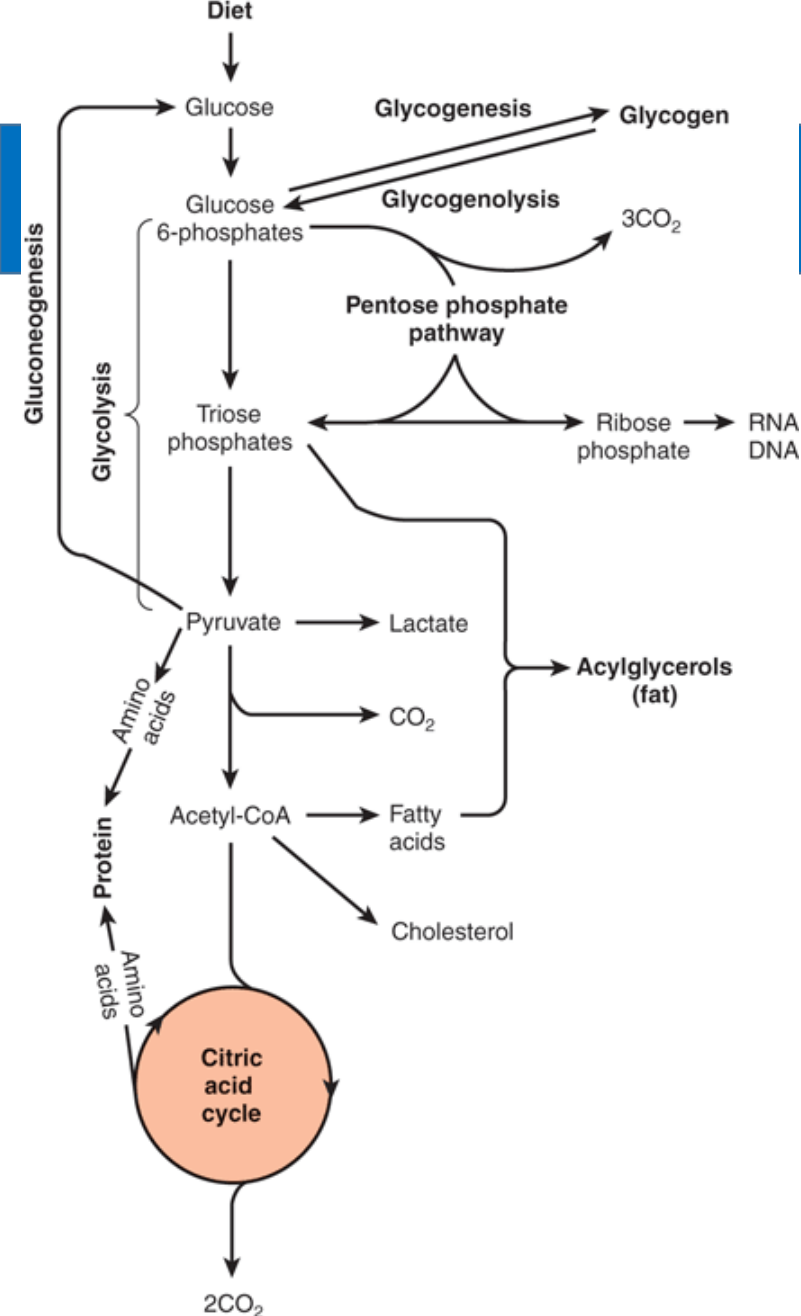


Utilisation of Carbohydrates

- (2) Anaerobic / aerobic respiration:

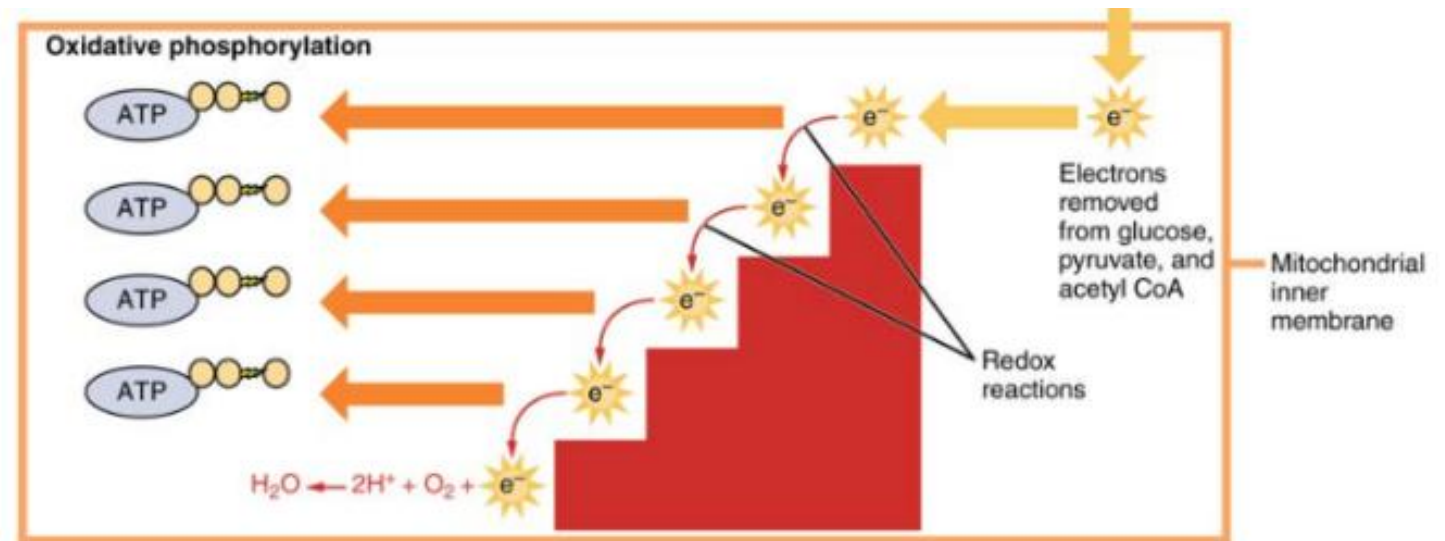
- During aerobic respiration (presence of oxygen), pyruvate enters the Krebs's cycle and **electrons are transferred** from pyruvate to receptors NAD^+ , FAD and ADP
- High energy molecules such as ATP, NADH, FADH_2 are created
- Acetyl CoA is formed in the process and is responsible for **fatty acid synthesis**
- Each Krebs's cycle produces **ONE ATP, ONE FADH_2 , and THREE NADH**
- Krebs's cycle also provide the starting materials (amino acids) for protein synthesis



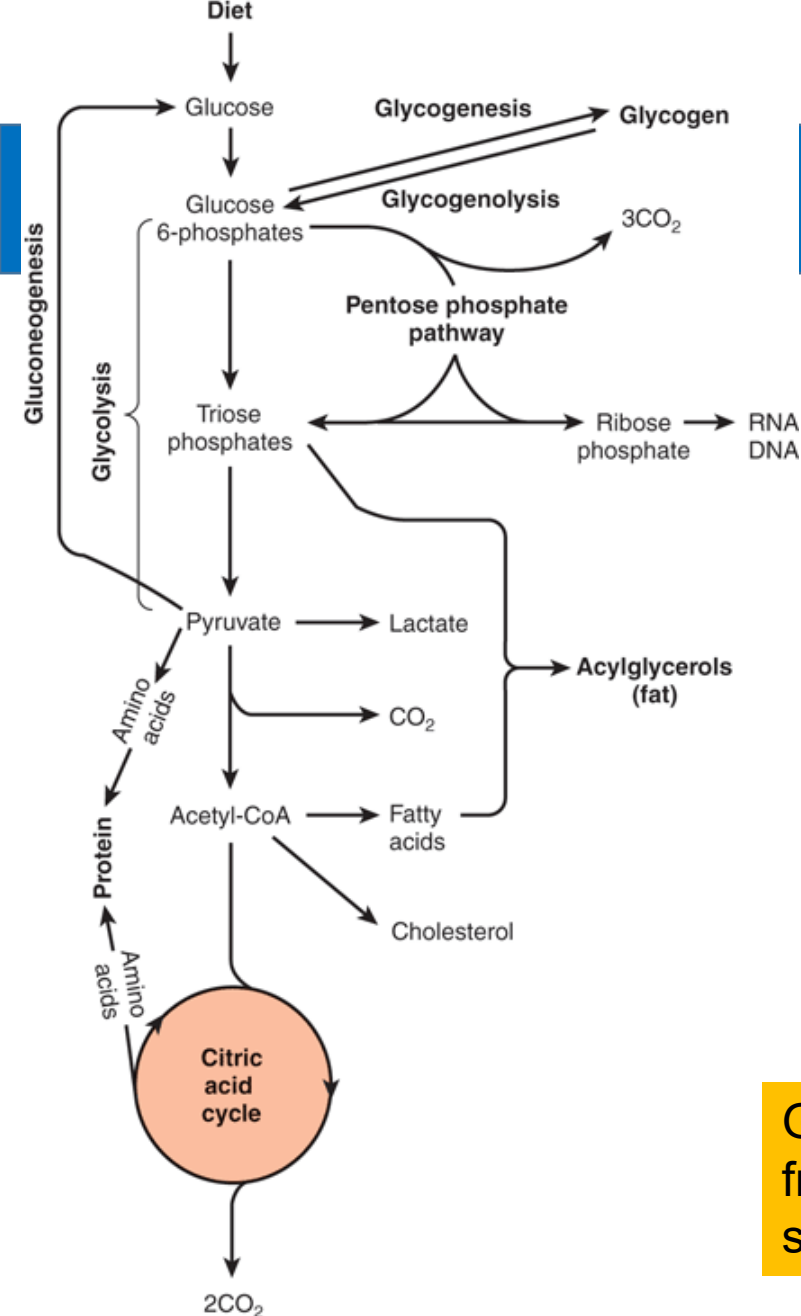


Utilisation of Carbohydrates

- (3) Oxidative Phosphorylation and Electron Transport Chain:
 - Oxygen supply is very essential for this process. Without oxygen electron flow ceases
 - ATP is produced in the process
 - **Oxygen is the terminal electron acceptor**
 - New water molecules are formed at the end of this process



For every glucose molecule, 30 ATPs are produced

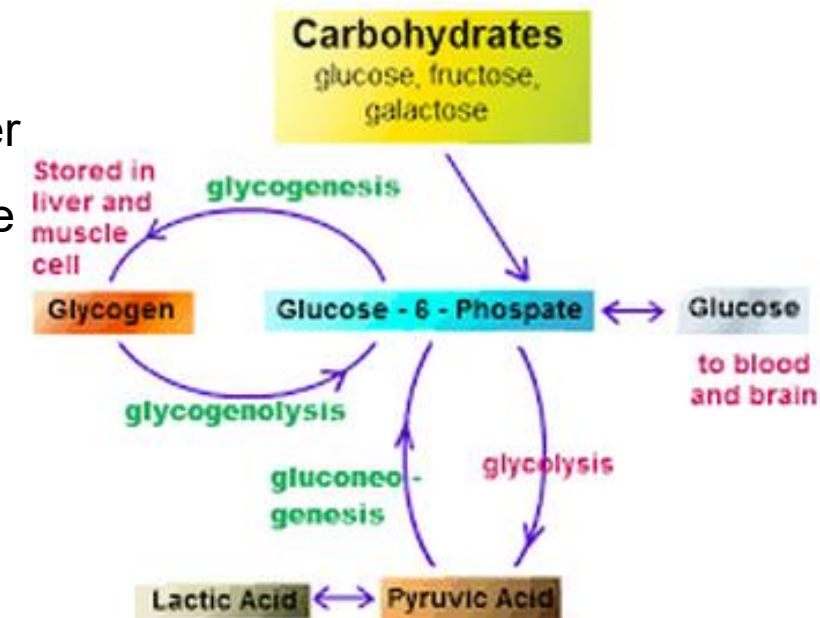


Utilisation of Carbohydrates

• (4) Gluconeogenesis:

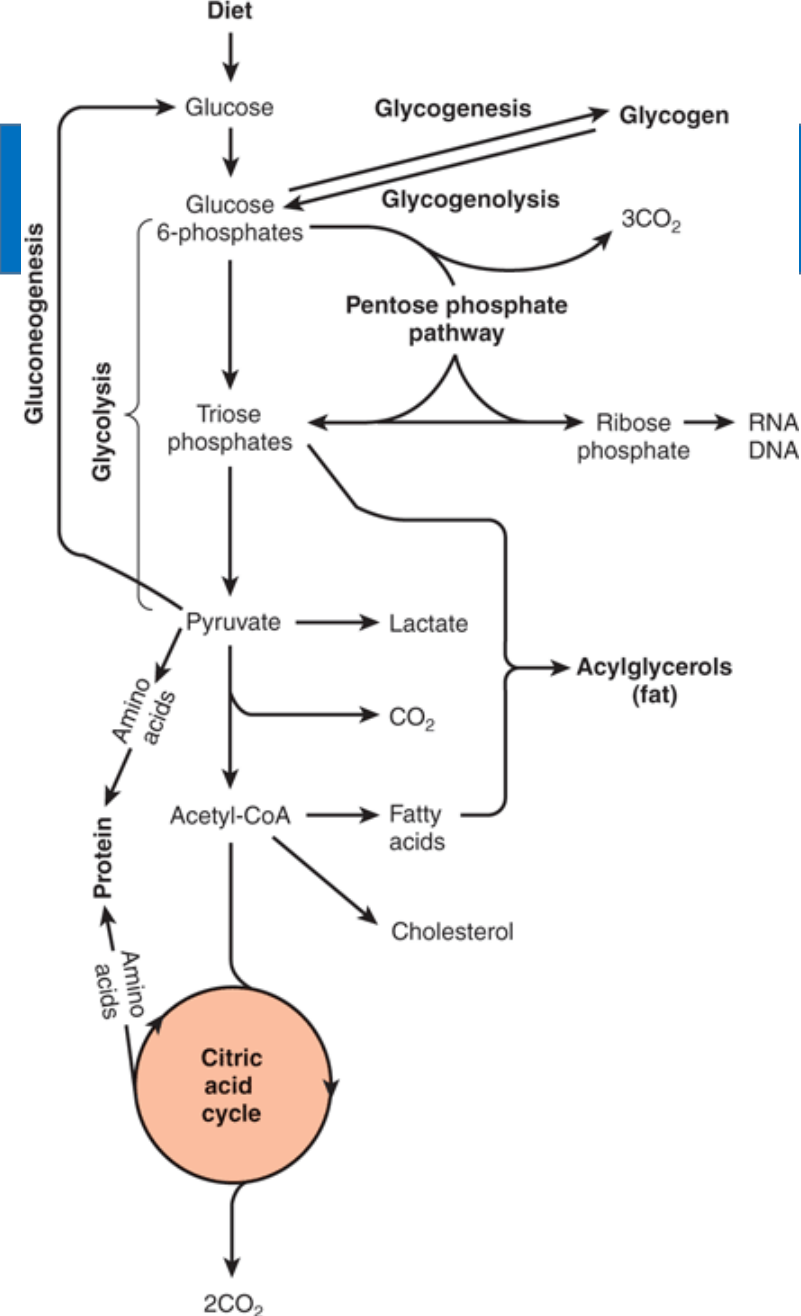
- Synthesis of new glucose molecules from pyruvate, lactate, glycerol, or the amino acids alanine or glutamine (from non – carbohydrate sources)
- Occurs during starvation, fasting or low-carbohydrate diets
- Gluconeogenesis is not just the reverse of glycolysis
- Occurs in the **liver**
- They are independent of each other
- This is an energy consuming phase

Glucose is the main monosaccharide from which other monosaccharides are synthesised



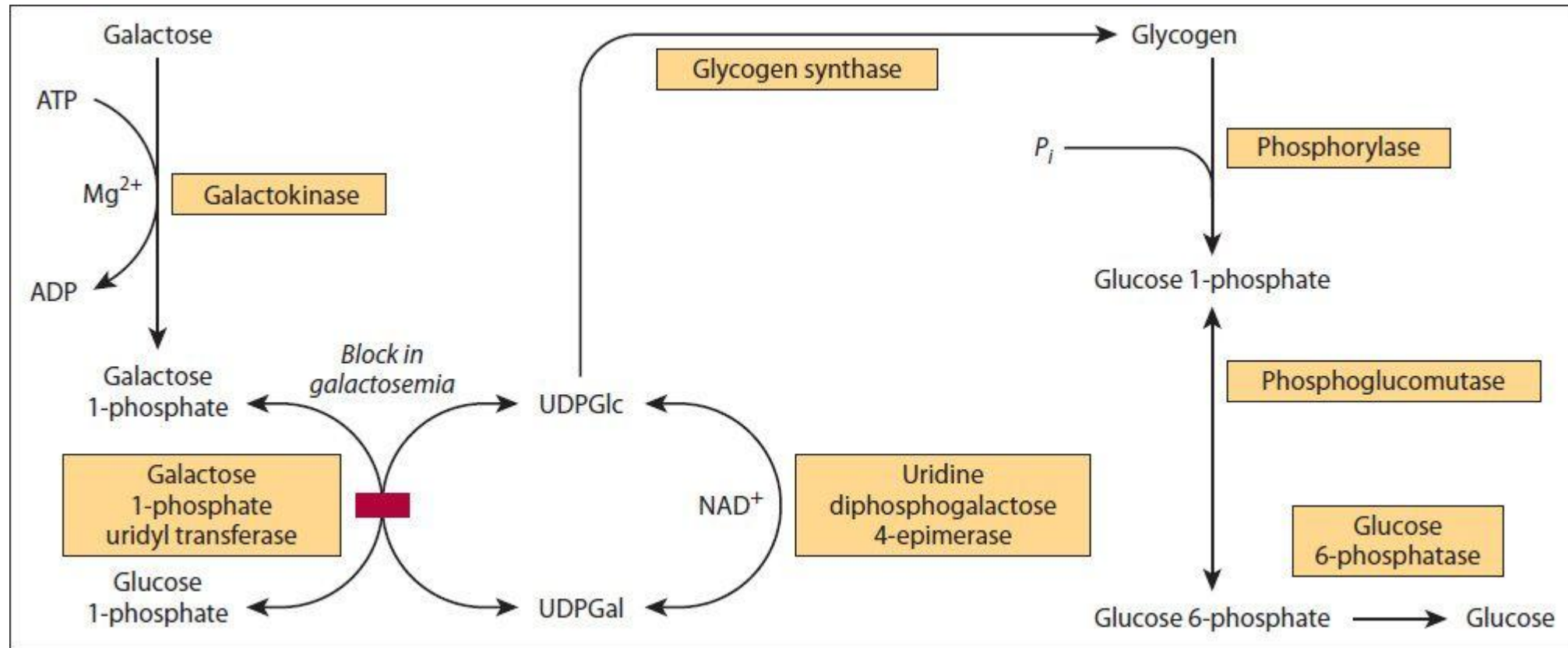
Utilisation of Carbohydrates

- (5) Glycogenesis:
 - Process of glycogen synthesis
 - Glycogen is a polymeric form of glucose
 - This is for storage
 - Activated during rest period **in liver**
- (6) Glycogenolysis:
 - Break down of glycogen to glucose and glucose-6-phosphate
 - Takes place in hepatocytes (liver) and myocytes (muscles)
- (7) Pentose phosphate pathway is responsible for the synthesis of starting molecules for the synthesis of nucleic acids



Utilisation of Carbohydrates

- Fate of other monosaccharides (eg. galactose)



This highlights the possibility of interconversion of molecules